

Answer Key — Biology Grade 11

Sub-Strand 3.3: Excretion and Homeostasis in Animals • Lesson 3: Following the Fluid: How the Nephron Filters Blood and Makes Urine

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	A	B	A	D	A	A

USE AFTER: OBSERVE PHASE — Clip 1: blood flow into the glomerulus and ultrafiltration into Bowman's capsule

Q1. Blood flows into the glomerulus in a mammalian kidney. In which part of the nephron does ultrafiltration take place?

- A. Proximal convoluted tubule, where glucose is taken back
- B. Collecting duct, where ADH controls water movement
- C. Loop of Henle, where the medulla is made concentrated
- D. Glomerulus and Bowman's capsule, where fluid is pushed out of the blood**

Correct answer: D. Ultrafiltration happens between the glomerulus and Bowman's capsule, where high blood pressure forces small molecules out of the blood. The other regions carry out reabsorption, concentrating or ADH-controlled water reabsorption.

USE AFTER: OBSERVE PHASE — Teacher pauses the clip at the glomerulus: 'Why are blood cells and large proteins not in the filtrate?'

Q2. Blood cells and large proteins are found in the blood entering the glomerulus but not in the glomerular filtrate. What is the best explanation?

- A. They are too large to pass through the filtering membrane under pressure**
- B. They are secreted into the distal tubule and leave in urine
- C. They are broken down by ADH inside Bowman's capsule
- D. They are filtered out first but are reabsorbed in the proximal tubule

Correct answer: A. Ultrafiltration is pressure-driven and depends on size, so only small molecules such as water, glucose, urea and salts pass into the capsule. Blood cells and large proteins stay in the blood.

USE AFTER: OBSERVE PHASE — 'Follow the Fluid' table: glucose after the proximal tubule (Clip 2)

Q3. A healthy person eats ugali and their blood glucose rises. Glucose is filtered into the nephron, yet none is found in the urine. What happens to the glucose?

- A. It is carried to the bladder but is too dilute to detect
- B. It is all reabsorbed into the blood in the proximal tubule using carrier proteins and energy**
- C. It is changed into urea in the loop of Henle
- D. It is destroyed in Bowman's capsule before reaching the tubule

Correct answer: B. Glucose is a useful substance, so it is completely reabsorbed by carrier proteins in the proximal tubule, using energy. None is left to enter urine.

USE AFTER: OBSERVE PHASE — 'Follow the Fluid' table: comparing urea in filtrate and urine

Q4. In the 'Follow the Fluid' table, urea is present in the glomerular filtrate and is much more concentrated in urine. Why is urea more concentrated in urine than in the filtrate?

- A. Water is reabsorbed along the nephron, but most urea stays in the tubule**
- B. Urea is reabsorbed into the blood faster than water
- C. ADH changes the water into urea
- D. Extra urea is made in the collecting duct

Correct answer: A. Much of the water in the filtrate returns to the blood, but urea is mostly not reabsorbed. The same amount of urea in less water makes the urine concentrated.

USE AFTER: EXPLAIN PHASE — Claim-evidence-reasoning: why the goat produces concentrated urine, and linking ADH to the panting dog

Q5. On a hot day a goat loses water through sweating and has not drunk much. Its brain releases more ADH. What does this cause in the kidney?

- A. The proximal tubule stops reabsorbing glucose
- B. The collecting duct reabsorbs less water, so a large volume of dilute urine is produced
- C. The glomerulus filters more proteins into Bowman's capsule
- D. The collecting duct reabsorbs more water, so a small volume of concentrated urine is produced**

Correct answer: D. ADH makes the collecting duct reabsorb more water into the blood. This saves body water and leaves a small volume of concentrated urine.

USE AFTER: EXPLAIN PHASE — Teacher prompt: 'Fish take in water by osmosis. What would their kidneys need to do differently?'

Q6. A freshwater fish takes in a lot of water by osmosis through its body surface. Which description of its kidney activity fits its dilute urine?

- A. It excretes the excess water as a large volume of dilute urine**
- B. It stops filtering blood so that no water enters the nephron
- C. It makes urine only when it is out of the water
- D. It reabsorbs almost all the water, so the urine is concentrated

Correct answer: A. Water floods into the fish, so its kidneys must remove the excess water. The result is a large volume of dilute urine, the opposite of the goat's water-saving urine.

USE AFTER: MODEL BUILDING PHASE — Revising the class model: 'Where on your model is the pressure? Where is the energy-using step?'

Q7. Which statement correctly contrasts ultrafiltration with selective reabsorption in the nephron?

- A. Ultrafiltration is pressure-driven and depends on size, while selective reabsorption returns useful substances using carrier proteins and energy**
- B. Ultrafiltration uses carrier proteins and energy, while selective reabsorption is driven by blood pressure
- C. Ultrafiltration happens in the collecting duct, while selective reabsorption happens in the glomerulus
- D. Both processes are driven only by blood pressure and work in the same part of the nephron

Correct answer: A. Ultrafiltration in the glomerulus is driven by high pressure and depends on molecule size. Selective reabsorption, for example of glucose in the proximal tubule, uses carrier proteins and energy to return useful substances to the blood.